
Verifying Financial Claims with On-Device Models

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Abstract

Financial claim verification requires validating short claims against long regulatory filings, making cloud-only inference costly and often unsuitable for sensitive documents. We build a routed edge–cloud pipeline that runs two small models on device and escalates only when a disagreement gate or an arithmetic trigger fires, with no training or fine-tuning. On 1,700 held-out claims from FINDVER, the routed system reaches 75.8% accuracy versus 77.4% for cloud-only inference, with no significant overall difference, while using 56.6% of the cloud tokens and settling 46.6% of claims entirely on device. That aggregate tie hides a substantial loss on one subset: agreement between two models of the same family becomes false confidence because their errors are correlated, costing 8.7 points on the information-extraction claims the gate keeps local. Arithmetic claims share this failure mode, since both local models fail in the same direction, and an unconditional trigger succeeds where a plain disagreement gate cannot. Two local rechecks that address both cases raise accuracy to 76.8% and cut escalation to 44.9%. Effective edge–cloud verification depends not only on local model accuracy, but on whether the routing signal can observe the failures that matter.

1 Introduction

Financial claim verification often requires grounding short claims in long, heterogeneous regulatory filings, documents that run to hundreds of pages [Reddy et al., 2024]. Accuracy remains low even for frontier systems: given a retrieval system over such filings, GPT-4-Turbo answered 81% of a manually reviewed sample of questions incorrectly or refused to answer [Islam et al., 2023]. Sending every filing to a frontier cloud model is a poor deployment fit for two reasons: filings may be commercially sensitive, and most input tokens come from the document rather than the claim, so repeated verification is expensive. Local models reduce both exposure and cost, but deciding when to trust them is difficult. The usual test is agreement: two local models read the same claim, and only a disagreement between them sends it to the cloud. That test assumes the two models fail independently, and some systematic failures produce no disagreement at all.

We address this problem with a routed edge–cloud verification pipeline that requires no training or fine-tuning. Retrieval and local inference remain on device, while a gate escalates only selected claims to a frontier model. We evaluate this design on FINDVER [Zhao et al., 2024], where each claim must be judged *entailed* or *refuted* against the SEC filing it refers to. The strongest published system on that benchmark reaches 75.0% under retrieval augmentation, and human experts reach 93.3%.

This paper makes three contributions. We measure the accuracy cost of a routed pipeline against a frontier cloud model, and find no significant difference at 53.4% of the cloud calls. We show that the aggregate result is misleading, because a gate built on agreement between two models of one family reads correlated error [Kim et al., 2025] as confidence, and we quantify what that costs on the subset

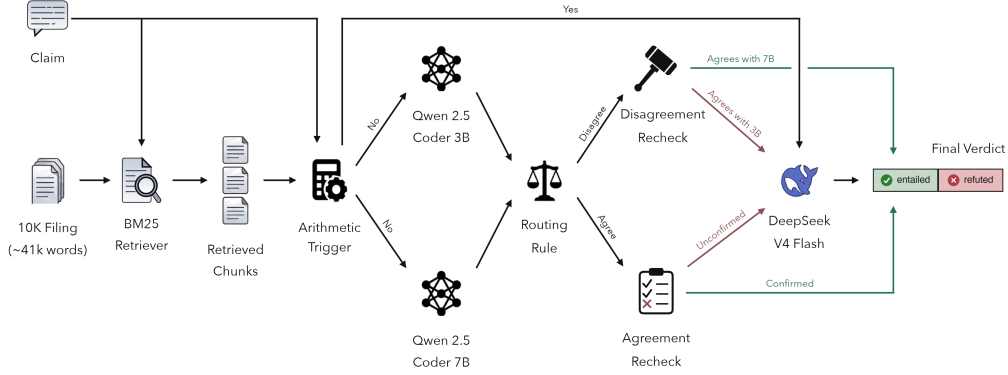


Figure 1: The pipeline. Retrieval is executed in Python with no model calls. Two small models on the device read the same retrieved evidence and answer independently. The gate sends a disagreement to the disagreement recheck and an agreement on *entailed* to the agreement recheck (Section 3.2). Green arrows from those two components keep the claim on the device and red arrows send it to the cloud, which is reached on 44.9% of claims. A claim asserting a computed quantity is escalated by the arithmetic trigger before either local model runs.

where it happens. We then address the two failures the gate cannot see, agreement on a wrong verdict and arithmetic error the two models share, on the device.

2 Related Work

FINDVER [Zhao et al., 2024] verifies claims against long SEC filings and separates them into information extraction (FDV-IE), numerical reasoning (FDV-MATH), and domain-knowledge (FDV-KNOW) subsets; because these documents exceed consumer-model context limits, evaluation typically uses fixed retrieval augmentation [Lewis et al., 2020]. The only other method we found evaluated on FINDVER is MACE [Saha et al., 2026], which decomposes verification into three roles but uses the same large model for all of them; its smallest reported configuration holds roughly 27B of parameters resident, where our two local models hold 10B between them; we instead ask which model should perform each role under consumer-device constraints. For numerical reasoning, Program of Thoughts [Chen et al., 2022], PAL [Gao et al., 2022] and TART [Lu et al., 2025] delegate the arithmetic to code instead of performing it in-model [Wei et al., 2022]; Section 5.3 shows why that fails at our model scale.

Small-to-large routing is commonly formulated as a cascade [Moslem and Kelleher, 2026], including in edge-cloud systems [Li et al., 2025], with deferral based on uncertainty [Kotte, 2026] or model agreement [Kolawole et al., 2025, Soiffer et al., 2025]. However, model errors are correlated [Kim et al., 2025], and even multiple frontier judges may provide few independent signals [Kohli, 2026]; Section 5.2 examines this issue in our two-model gate.

Financial claim verification has itself moved quickly since FINDVER appeared, in FinGround [Guo et al., 2026], VeriFin [Hall et al., 2026] and FinVerBench [Panda, 2026]. None of the three is evaluated on FINDVER, and none targets a device that cannot hold a frontier model, which is the constraint that shapes our design.

3 Method

Given a claim and its regulatory filing, our system retrieves relevant evidence, attempts verification locally, and escalates only when the local tier is not trusted (Figure 1). Retrieval and local inference run entirely on device; the cloud model is an exception path rather than the default. No component is trained or fine-tuned.

66 3.1 On-Device Retrieval and Inference

67 We retrieve the top $k=10$ filing elements with BM25 [Robertson and Zaragoza, 2009]. The retriever
68 uses no neural model or API call. The retrieved evidence is then given independently to Qwen2.5-
69 Coder models with 3B and 7B parameters, both quantized to four bits and served through Ollama.
70 They receive the same prompt and evidence, use a 32,768-token context window with temperature
71 zero and a fixed seed, and do not observe each other’s outputs.

72 The two models belong to the same family and were selected for deployment efficiency rather than
73 diversity. This makes their agreement inexpensive to obtain but, as we show in Section 5.2, not
74 always a reliable confidence signal.

75 3.2 Routing Policy

76 Routing uses three signals. An arithmetic trigger, written in plain Python with no model behind it,
77 identifies arithmetic claims before local inference and escalates them. The gate escalates any claim
78 on which the two local models disagree. We report those two together, without the rechecks, as *gate*
79 *only*. An agreement recheck then examines selected agreements, and a disagreement recheck revisits
80 disagreements; both run on the device and each costs at most one further 7B call. None of the three
81 produces a verdict of its own; each only decides whether the claim is kept locally or escalated to the
82 cloud. Whenever a claim is escalated, the cloud verdict replaces the local verdict.

83 **Arithmetic Trigger.** A fixed rule-based detector identifies claims that assert a computed quantity
84 from the claim text alone. It uses three regular-expression patterns covering sentence-final numeric
85 predicates, percentage claims, and approximate or exact numeric assertions. Detected claims are sent
86 directly to the cloud. This trigger targets failures that disagreement cannot expose when both local
87 models make the same arithmetic error.

88 **Agreement Recheck.** When both local models predict *entailed*, we apply two checks for missed
89 contradictions. First, a deterministic scan tests whether numerical values asserted by the claim occur
90 anywhere in the filing under common formatting and scale variants. It is the only part of the system
91 that reads outside the retrieved passages, and it runs no model and returns no verdict, so every verdict
92 still comes from retrieved evidence alone. Second, the 7B model independently checks the retrieved
93 evidence for an unsupported detail. Either signal triggers escalation; otherwise the shared local
94 verdict is retained.

95 **Disagreement Recheck.** For non-arithmetic claims on which the local models disagree, the 7B
96 model receives one additional query over the same evidence. Rather than asking it to choose between
97 the previous verdicts, the prompt asks independently for evidence contradicting the claim. If the
98 resulting verdict agrees with the 7B model’s original answer, that answer is retained locally; otherwise
99 the claim is escalated.

100 The two rechecks act on disjoint populations. Together with the arithmetic trigger, they separate three
101 cases that a single disagreement gate conflates: shared systematic errors, informative disagreements,
102 and disagreements that the stronger local model can resolve.

103 4 Experimental Setup

104 FINDVER provides no training data and our system has no trainable parameters, so its 700-claim
105 development split is used only to select the retriever, k , local models, prompts and routing signals,
106 and every figure is measured on the 1,700-claim test split. We report verdict accuracy under strict
107 scoring, where a response with no recoverable verdict counts as incorrect, with the unparseable rate
108 beside it. All latency measurements come from a consumer-GPU desktop. Appendix A.2 gives the
109 scoring procedure, the configuration and the hardware in full.

Table 1: Accuracy on 1,700 held-out claims, strict scoring. The lower block is the fraction of each subset that reached the cloud. Subsets are 600, 600 and 500 claims. Routing adds 8.0 points over the stronger local model alone. Neither routed system differs significantly from the cloud-only baseline, at $p = 0.119$ for the gate alone and $p = 0.583$ with both rechecks.

SYSTEM	OVERALL	FDV-IE	FDV-MATH	FDV-KNOW
3B alone	59.8%	63.3%	57.3%	58.4%
7B alone	67.8%	68.5%	66.5%	68.6%
routed, gate only	75.8%	77.5%	77.3%	72.0%
routed, + both rechecks	76.8%	78.8%	77.3%	73.6%
cloud only	77.4%	82.3%	77.2%	71.6%
escalated, gate only	53.4%	36.5%	79.8%	42.0%
escalated, + both components	44.9%	23.2%	80.5%	28.4%

Table 2: Cost and latency over the same 1,700 held-out claims. Token counts are recorded per claim. Latency is the GPU desktop; see Section 4.

	GATE ONLY	FULL SYSTEM	CLOUD ONLY
cloud calls made	908	764	1,700
cloud input tokens	3,400,854	2,890,886	6,319,626
cloud output tokens	2,106,383	2,041,975	3,560,933
cloud cost	\$1.066	\$0.977	\$1.882
claims answered on device	46.6%	55.1%	0%
median latency per claim	26.0 s	26.7 s	8.0 s
95th percentile latency	86.8 s	87.1 s	71.8 s

5 Results

5.1 Routing matches cloud accuracy with fewer cloud calls

With the gate alone the routed system reaches 75.8% accuracy against 77.4% for cloud-only inference (Table 1). The difference is not significant under paired McNemar testing ($p = 0.119$), and that configuration was fixed before the test split was read.

Adding the two rechecks raises accuracy to 76.8% ($p = 0.583$) and reduces escalation from 53.4% to 44.9%, leaving 55.1% of claims entirely on device (Table 2). Both were designed after output from this split had been read, so we report them as a separate row (Appendix A.1).

The cost reduction does not extend to latency. Both local models must run before most routing decisions can be made, so median end-to-end latency is 26.0 s for the routed pipeline versus 8.0 s for cloud-only inference.

5.2 Agreement can hide correlated error

The aggregate result masks a failure of the basic disagreement gate. Table 1 shows that the gate-only system is significantly worse than the cloud baseline on FDV-IE, while the two systems are statistically tied on FDV-MATH and FDV-KNOW.

The deficit comes from claims the gate keeps locally. On the 381 FDV-IE claims retained on device, the routed system scores 76.9%, whereas the cloud model would score 85.6% on the same claims, a gap of 8.7 points. Performance on escalated FDV-IE claims is comparable to the cloud baseline, so the problem is not insufficient escalation overall but misplaced confidence in a particular set of agreements.

Because the two models share an architecture, a tokenizer and much of their training distribution, their errors are correlated: when both make the same mistake, agreement suppresses escalation exactly as if both were correct. Disagreement is useful only to the extent that the models fail differently.

Table 3: The local and cloud verdicts on the claims each routing decision acts on. *Local* is the verdict accuracy the device would have returned and *cloud* is the verdict accuracy the cloud model returned or would have returned, scored on the same claims. The first three rows are escalations. The fourth is a retention, where the disagreement recheck keeps the claim on the device.

ROUTING DECISION	CLAIMS	LOCAL	CLOUD
escalate, the two models disagree	459	57.3%	73.4%
escalate, arithmetic claim	449	64.4%	78.2%
escalate, agreement recheck fires	102	57.8%	66.7%
keep, disagreement recheck sides with the 7B	269	74.7%	73.2%

133 The agreement recheck targets this failure mode. With the complete system, FDV-IE accuracy rises
 134 from 77.5% to 78.8%, and the difference from the cloud baseline is no longer significant ($p = 0.055$).
 135 On locally retained FDV-IE claims, the gap to cloud performance narrows from 8.7 to 4.1 points.

136 5.3 The routing signals are complementary

137 Over all 908 claims the gate escalated (Table 3), the cloud model scores 75.8% against 60.8% for
 138 the 7B model had those claims been kept locally. The two original signals capture different errors:
 139 the arithmetic trigger operates on claims where the local models may agree, and so detects failures
 140 disagreement cannot observe.

141 We also tested executing arithmetic locally rather than escalating it, following the Program-of-
 142 Thoughts pattern [Chen et al., 2022]. The approach failed at operand selection, not at execution:
 143 supplied with the correct operands the same deterministic interpreter reaches 91.2%, but small
 144 local models often select the wrong figures from evidence containing many numerical candidates.
 145 Increasing the extractor from 3B to 7B narrows the gap to the cloud model without closing it, so we
 146 use arithmetic detection as a routing signal and not as a local solver.

147 Of the 351 claims where the two local models agreed on *entailed*, the agreement recheck passes
 148 claims that are correct 83.5% of the time and flags the rest. The disagreement recheck keeps 269 of
 149 482 non-arithmetic disagreements on the device at no cost in accuracy, and its signal is asymmetric:
 150 the 7B verdict is correct 74.7% of the time when the recheck agrees with it, against 33.7% when it
 151 does not.

152 5.4 Missing evidence dominates the remaining errors

153 A single failure mode accounts for 63.4% of knowledge-subset errors and 41.0% of all system errors,
 154 at a similar rate on the development split: the model states that the required evidence is absent and
 155 refutes the claim on that basis. On the held-out knowledge subset, 38.4% of responses invoke missing
 156 information, 98.4% of those then predict *refuted*, and 47.6% of those predictions are incorrect. The
 157 same pattern produces a broader label bias. Although 50.1% of gold labels are *entailed*, the cloud
 158 model predicts that label on only 33.9% of claims and the 3B model on only 39.5%, and the routed
 159 system partially reduces the imbalance: evidence the model fails to locate becomes a refutation
 160 instead of an abstention.

161 Finally, an oracle that selects the correct verdict whenever either local model produces it would reach
 162 79.9% without any cloud calls; allowing selection among both local models and the cloud model
 163 raises the bound to 84.8%. Neither is implementable, since each needs the gold label in order to
 164 select, but they show that a substantial fraction of the remaining error is attributable to routing and
 165 not to the absence of a correct candidate verdict.

166 6 Conclusion

167 We show that selective edge-cloud routing can match cloud-only accuracy on financial claim verifica-
 168 tion while substantially reducing cloud use. The limit is not local model accuracy but the routing
 169 signal: agreement between related models can become false confidence, and some errors never
 170 produce disagreement at all. A cascade needs a signal built around the structure of its local errors,
 171 not agreement alone.

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226 A Appendix

227 A.1 Limitations

228 **Design and evaluation on the same split.** The FDV-IE deficit and its mechanism were identified by
229 reading the output of the held-out run reported in Section 5.2, and the two components of Section 3.2
230 were designed in response to it and then evaluated on that same split. They therefore do not carry
231 the development-split protocol that the rest of the system follows, and we report them as a separate
232 system row in Table 1 rather than folding them into the gate-only arm. The gate configuration reported
233 there was frozen before that split was read and is unaffected. Both components were also evaluated on
234 the development split. The cost reduction reproduces there almost exactly, from 54.0% to 42.0% of
235 claims escalated against 53.4% to 44.9% here. Accuracy on that split is flat, 79.0% with and without
236 the components, with FDV-IE at 83.6% against 83.2%; that split escalates a comparable fraction of
237 claims but its disagreement population is a harder one, and we report the figure rather than resting the
238 components’ case on it.

239 **One benchmark, one document type.** Every result here comes from FINDVER, which is built on
240 English-language SEC filings. Neither the retriever’s advantage nor the arithmetic trigger’s precision
241 should be assumed to transfer to another document class, and the detector in particular exploits a
242 regularity of phrasing that may be an artefact of how these claims were written.

243 **Reproducibility.** The cloud endpoint does not honour a temperature or a seed, so its verdicts vary
244 between identical runs. Our own local models are seeded at temperature zero and still move roughly
245 one verdict in ten between reruns on the same machine. Comparisons in this paper are therefore
246 paired wherever a paired test is available.

247 A.2 Detailed Experimental Settings

248 **Splits and protocol.** FINDVER provides a development split of 700 claims and a test split of 1,700
249 claims, both with labels. The benchmark ships no training data, and no component of this system has
250 trained parameters, so development consists of selecting components and settings rather than fitting
251 them. We carry out that selection on the development split, including the retriever, the value of k , the
252 choice of local models, the prompts and both gate triggers, and we report all figures on the test split.
253 Reruns of an identical configuration on this hardware move roughly one verdict in ten, so we treat
254 any difference smaller than that as noise rather than as a result.

255 **Scoring.** A verdict is extracted from the model’s response by a rule-based extractor that searches the
256 whole response, not only its final sentence, since small models frequently place the verdict elsewhere.
257 Responses from which no verdict can be recovered are counted in an explicit unparseable category
258 and scored as incorrect, which we call strict scoring. We report the unparseable rate alongside
259 every accuracy figure, because a system that fails to answer and a system that answers wrongly are
260 different failures and strict scoring alone conflates them. On the held-out split the rate is 1.65%
261 for the 3B model, 2.82% for the 7B, and 0.94% for the routed system. Where a figure of ours is
262 placed beside a published one, we additionally report it under the benchmark’s own convention, in
263 which unrecoverable verdicts are resolved by a coin flip. Comparing a strictly scored number with a
264 coin-flipped one would understate ours by exactly the amount our models fail the output format.

Table 4: Macro recall at $k=10$, the metric the benchmark’s own evaluation script computes. Our BM25 replicates to within 0.6 points on the held-out split it was never tuned on, and beats the strongest published retriever there by 5.4 points with no neural model and no API cost.

RETRIEVER	DEV($n=700$)	HELD-OUT($n=1,700$)
ours, BM25	74.60%	75.16%
text-embedding-3-large	68.01%	69.75%
benchmark’s BM25	65.16%	64.29%
contriever-msmarco	33.48%	35.02%

Hardware. Experiments were executed on a desktop with a consumer GPU, since a full run of the routed pipeline over the test split takes about 15.8 hours there. The device of record for the deployment claim, however, is a 2017 laptop with 16 GB of memory and no usable GPU, and every latency figure must name its machine because the ordering reverses between them. On the desktop the local 3B model is the fastest of the three at 6.9 seconds per claim, faster than either cloud model; on the laptop the same model takes roughly 420 seconds. Every latency figure in this paper comes from the desktop.

A.3 Retrieval

Our BM25 retriever reaches 75.16% macro recall at $k=10$ on the held-out split, 5.4 points above the strongest retriever previously reported on FINDVER (Table 4). The result uses no neural model and no API call.

The gain is largely an implementation effect rather than a new retrieval method. Development-set ablations attribute it mainly to the IDF formulation and punctuation handling; additional claim decomposition, dense fusion, and parameter sweeps do not improve the final retriever.

More importantly, retrieval recall is only weakly coupled to downstream verification in our pipeline. Supplying perfect evidence does not significantly improve accuracy ($p = 0.116$), while increasing k from 10 to 20 substantially raises recall but produces no measurable accuracy gain despite increasing prompt length. We therefore report the BM25 result as a retrieval result, not as an explanation for the verification gains above.

A.4 Future Work

A verification pipeline whose default path runs on consumer hardware ties a frontier cloud model on 1,700 held-out financial claims, at 51.9% of the cost, with 55.1% of filings never leaving the device and with no training at any stage. The aggregate tie has structure, and identifying that structure was what produced the two components that close most of it.

The mechanism is specific and it is actionable. A gate built on agreement between two small models reads correlated error as confidence, and because our two local models come from one family their agreement on the information-extraction subset was worth 8.7 points on the claims the gate kept. The two components of Section 3.2 address the two sides of that failure directly: the agreement recheck re-examines the agreements the gate trusts, and the disagreement recheck resolves the ones it does not. Together they narrow the FDV-IE deficit on kept claims from 8.7 to 4.1 points, raise overall accuracy from 75.8% to 76.8%, and remove 269 cloud calls, so that the system answers a majority of claims without contacting the cloud at all. Both components are local, both cost at most one additional 7B call, and neither requires a second model family or any training.

The general lesson is that the value of a cascade gate is bounded by the independence of the signals it reads, and that when independence is unavailable it can be recovered cheaply by asking the same model a different question. The disagreement recheck does not add a model; it adds a question whose answer is informative precisely because it is not framed as a decision between two known positions.

Three further directions come out of the measurements above. The escalation triggers are hand-written, and the taxonomy in Section 5.4 suggests a learned or pattern-derived trigger aimed at the specific failure that dominates the error budget, which the present gate does not detect at all. A trigger of that kind is one half of an agentic skill in the sense now being formalised [Jiang et al., 2026], a

306 reusable procedure carrying its own applicability condition; the other half, the procedure itself, is
307 where the interpreter we did not build would sit. The oracle bound indicates that most of the remaining
308 gap is a selection problem and not a capability problem, so a better gate is a better investment than a
309 better local model. Every latency figure here comes from a desktop, so measuring the pipeline end
310 to end on a CPU-only consumer machine is necessary before any claim about deployability on that
311 class of hardware, and we make none here.